

Every series is three things at once

The shared vocabulary between the data scientist and the decision-maker.

$$y_t = T_t + S_t + R_t \quad y_t = T_t \times S_t \times R_t$$

additive

multiplicative $\Leftrightarrow \log y = \log T + \log S + \log R$

T_t trend

the long-run level — growth, decline, a structural break

S_t seasonality

the repeating pattern of KNOWN period: 24, 168, 8766

R_t remainder

what neither explains — and where a missing covariate hides

**Decide on EVIDENCE: correlate the per-cycle level with the per-cycle amplitude.
If the swing grows with the level, the series is multiplicative — model $\log(y)$.**